

# Trust Without Understanding: A Case Study of Industrial Computer Vision in Protein Processing

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## Abstract

Trust in AI systems is often treated as contingent on user understanding and model explainability. This paper examines how organizational trust developed around a computer vision system deployed in legacy protein processing facilities where digital infrastructure was historically minimal. Drawing on nine semi-structured interviews with staff at the technology provider and the processor's internal project lead, we analyze how the system became accepted not as "AI," but as measurement infrastructure embedded within existing quality control regimes. Organizational trust developed through hardware reliability in a harsh environment, validation rituals grounded in existing quality frameworks (e.g., ANOVA, gauge R&R), low-stakes framing of error, and the system's fit with established organizational categories and routines. We show how ground truth and "accuracy" were pragmatically adjusted to preserve usability, and argue that trust in industrial AI is an organizational and infrastructural accomplishment, challenging assumptions that explainability is a universal prerequisite for trust.

## CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; Collaborative and social computing; • **Social and professional topics** → Computing in organizations; • **Computing methodologies** → Artificial intelligence.

## Keywords

Trust in AI, Trust in automation, Human-centered AI, Industrial AI, Qualitative interviews, Sociotechnical systems, Organizational practices

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## 1 Introduction

In many protein processing facilities, production practices have changed little over the past century. Carcasses move along conveyor belts and overhead rails, inspected and trimmed by workers

using knives, with limited digital instrumentation embedded in core workflows. This differs from other industrial sectors, such as automotive and pharmaceutical manufacturing, where artificial intelligence and automation are more deeply embedded in core production processes [7]. While trust in automation has been widely studied in domains such as aviation and advanced manufacturing, less is known about how AI is introduced into legacy industrial settings with minimal digital infrastructure. In such environments, AI can fundamentally reshape how work is monitored and evaluated. This paper examines how trust in an AI-based computer vision (CV) system emerges at the organizational level as the system becomes embedded within legacy workflows.

Research on trust in AI has emphasized concerns such as explainability, transparency, and bias, often framing trust as a function of users' understanding of algorithmic decision-making [3]. At the same time, trust in automation research conceptualizes trust as calibrated reliance within dynamic human-machine systems [11]. Lee et al. emphasizes that trust is a dynamic, history-dependent variable that must be precisely calibrated to the system's actual capabilities to ensure appropriate reliance. Within this framework, trust evolution is viewed as a closed-loop process shaped not only by direct experience with the technology but also by the broader individual, organizational, and cultural contexts in which the operator functions [11]. However, less attention has been paid to how trust is constructed at the organizational level, particularly before frontline workers directly interact with a system. Sociotechnical and practice-based accounts of technology use suggest that trust is produced not only through individual belief but through routine, institutional endorsement, and processes of breakdown and repair [17, 18]. From this perspective, technologies become trusted as they stabilize into everyday practice and recede into infrastructure. In industrial contexts where plants operate with a degree of autonomy, corporate-level decisions to adopt and mandate new technologies play a critical role in shaping downstream reliance. How corporations decide to trust AI systems prior to widespread deployment across sites is comparatively underexplored.

While CV systems are increasingly developed for manufacturing and food processing, much of this work emphasizes technical performance rather than the sociotechnical dynamics of implementation and adoption in active plants [13]. Research on AI adoption has expanded within healthcare settings, the dynamics of trust in these environments are shaped by high-risk decision-making, clinical accountability, and patient safety concerns [6]. By contrast, the computer vision system examined in this study operates within a lower-risk industrial context, where errors primarily affect yield optimization rather than life-or-death outcomes. Trust is context



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dependent, and assumptions derived from clinical AI cannot be directly transferred to industrial production environments. Understanding trust in low-risk, economically driven AI deployments requires attention to different organizational pressures and evaluative criteria.

The system under study is a fixed, camera-based CV system installed on a production line. It captures images of carcasses from multiple angles and produces near real-time yield-related measurements displayed through dashboards tailored to different organizational roles. The system does not directly intervene in production; it provides observational data intended to inform oversight and performance improvement. This positioning distinguishes it from autonomous or decision-making AI systems. Rather than replacing human judgment, it functions as a measurement technology embedded within existing managerial practices, offering a unique case for examining how AI becomes institutionalized as organizational infrastructure.

Through an interview study with technology-company staff and the protein processor's internal project lead, we show that organizational trust in the system did not emerge through understanding of AI mechanisms or demands for explainability. Instead, organizational level trust was built through hardware reliability, familiar validation practices, the low-stakes framing of error, and the integration of the system into existing organizational categories of measurement and performance evaluation. Interviews further indicate that stakeholders interpreted the AI-based system through mental models developed for prior automation technologies, treating it less as probabilistic AI and more as deterministic instrumentation. By examining organizational trust formation in a low-risk industrial context, this study contributes an empirical account of how corporations develop trust in new AI systems and challenges the assumption that explainability is a universal prerequisite for trust.

## 2 Methodology

We conducted a qualitative, semi-structured interview study to examine the adoption of an AI-based CV system in protein processing facilities. The study protocol was reviewed by the University of Notre Dame's Institutional Review Board and determined to be exempt from ongoing IRB oversight. All participants provided informed consent and participated voluntarily.

### 2.1 Study Context

The technology company installed its first pilot system with the large protein processor in 2024. Following the initial pilot, the deployment expanded across facilities, enabling retrospective examination of how trust was established as the technology and organizational practices co-evolved.

### 2.2 Participants and Data Collection

Participants were recruited from the company who built the CV system based on their involvement in developing, deploying, or supporting the system. They were recruited via internal communication channels by sharing details of the study. All eligible employees at the technology company volunteered to participate, resulting in seven interviews with roles spanning leadership, engineering, and field deployment. We also conducted two interviews with the

project lead at the protein processor, who coordinated evaluation, installation, and roll-out across facilities. Access to other staff from the protein processor such as frontline plant employees was limited due to organizational constraints and the sensitive nature of the production environment. To preserve anonymity during review, company names are withheld and additional participant details are omitted.

Interviews lasted 60 to 90 minutes and followed a semi-structured protocol covering the origin of the collaboration, early expectations, installation and troubleshooting experiences, validation practices, and moments of trust-building or trust erosion. Interviews also probed whether factors commonly associated with trustworthy AI (TAI) (e.g., explainability, privacy, fairness, accountability) were salient in adoption. The full interview protocol is included in the Appendix. Participants were not compensated.

## 2.3 Analysis

Interviews were audio-recorded and transcribed with participants consent using MacWhisper (Large v3 Turbo). Transcription accuracy was verified through review against original recordings, and transcripts were anonymized prior to analysis. We analyzed the data using reflexive thematic analysis following Braun and Clarke [4]. Analysis began with familiarization and inductive open coding on approximately 10% of the dataset to identify salient concepts. Through iterative discussion among the authors, codes were refined and applied to the full dataset, with the codebook evolving as analysis progressed. While coding was led by a single author, analytic rigor was supported through collaborative reflection consistent with reflexive thematic analysis. Codes were then clustered into higher-level themes capturing patterned meanings across participants and stages of deployment.

Representatives from the technology company were invited to provide feedback on the code definitions to support factual accuracy and domain understanding. They did not participate in data collection, coding decisions, theme development, or interpretation. The authors received no financial, material, or other support from the company.

The authors used an AI-based writing assistant to refine phrasing and organization of the manuscript. The tool was not used for data collection, coding, theme generation, or interpretation; all analytic decisions were made by the authors.

## 3 Findings

Our analysis shows that the participants believe protein processor staff's trust in the system was built through a combination of organizational need, hardware reliability and familiar validation practices. Across interviews, participants described how the protein processor consistently framed the system not as a probabilistic AI, but as a form of measurement infrastructure whose legitimacy rested on consistency, responsiveness, and fit with established practices. The participants believe that plant users' understanding of AI and its results did not played a role in building users' trust in the system. We organized our findings into five themes.

### 3.1 Risk Acceptance Driven by Unmet Operational Need

Participants described the system's adoption as occurring in a context shaped by prior failure and unmet operational need rather than enthusiasm for AI. Before working with the technology company, the protein processor had attempted both an internally developed solution and a third-party vendor system, neither of which was successfully adopted. These experiences generated skepticism within the protein processor plants and management, but also a sense of urgency. The internal project lead described the installation of the first device as a calculated risk taken in the absence of viable options. *"It was something that came about due to ... (the company) and the production plants being desperate for things to move the needle. Their key performance metric was in the absolute tank, and they were trying to do a shotgun approach to improve it."*

Participants emphasized that the pre-existing manual process was labor-intensive, difficult to standardize, vulnerable to cherry-picking, and poorly suited to the scale and pace of production. As a result, stakeholders were more willing to accept the risk of a new system as existing alternatives were perceived as inadequate.

### 3.2 Trust Built Through Hardware Reliability and Embodied Demonstration

Early trust in the system was strongly shaped by perceptions of hardware reliability. Participants consistently reported that initial skepticism from plant staff focused on whether the system could survive the harsh processing environment, such as exposure to water, temperature fluctuations, and debris, rather than on the validity of its measurements.

Interviewees described that on-site demonstrations, such as pressure-washing the enclosure to show it remained watertight, functioned as key trust-building moments for staff at the protein processor. Participants also cite engineers' responsiveness, including traveling on short notice to troubleshoot issues, further reinforced perceptions of accountability and ownership in the eyes of the protein processor staff. *"When we did that install I took the pressure washer and hosed it down in front of him and then checked it afterwards. I'm like, no, it's dry. It's fine. So that was a big shifting point for them... it was like an instant vote of confidence for us that the equipment's solid. So let's just make sure the data's good now."* In these accounts, trust emerged through visible material reliability rather than abstract assurances of accuracy. Concerns about measurement validity were secondary to establishing that the system could physically withstand the environment.

### 3.3 Trust Through Established Quality Frameworks : Positioning the System as a Measurement Instrument

Although the system relied on machine learning models, participants described that the protein processor's staff trust in the system was not tied to understanding how those models worked. Instead, participants reported that trust in the accuracy of the system was built through practices drawn from established manufacturing quality frameworks, particularly Six Sigma methods such as gauge repeatability and reproducibility and ANOVA testing.

According to participants, these practices did not function to interrogate probabilistic uncertainty or model behavior. Rather, they served to position the system, in the processor's view, as a measurement device comparable to thermometers or pressure gauges already in use. Participants noted that passing an ANOVA was frequently treated as the decisive criterion for trust, even when processor staff were understood to have limited familiarity with specific accuracy thresholds. The protein processor's internal project manager said *"I didn't trust the system until our first round of ANOVAs. That was the first that I actually trusted the system and I was incredibly surprised but I expected it to fail the first one"*. By embedding acceptance within familiar statistical rituals, the system aligned with existing epistemic standards, reducing pressure to engage with model internals or explainability and reinforcing its treatment as deterministic instrumentation rather than adaptive AI.

### 3.4 Definitions of Accuracy Adjusted and Error as Low-Stakes

Participants described multiple instances where the system was technically accurate while still producing results that felt "wrong" to experienced staff. *"They think that this is 30 grams but it turns out it's actually 60 or 80... A human can't tell the difference between 60 and 70 grams."* In such cases, adjustments were made to align outputs with human expectations rather than insisting on a single fixed ground truth. A participant explained that *"the allowance is only X grams and right off the bat at the first plant, we were showing the carcasses have 5X grams left... that's very demotivating for an operator... So [the protein processor] decided to bump the allowance up to 2X grams."*

These adjustments were supported by a low-stakes framing of error. Because the system did not directly intervene in production and because outputs were interpreted in aggregate over time, individual inaccuracies were viewed as tolerable. The internal project manager outlined that what mattered was stability across minutes, shifts, and lines rather than precision at the level of individual carcasses. *The reason why repeatability is so important is even if for this application precision is more important than accuracy. If we're off from true accurate value as long as we're consistently off and we're measuring a trend and then if they improve we show that improvement that's actually all that matters. They just need a consistent feedback signal.*

### 3.5 Explainability as Organizationally Irrelevant

Despite explicit probing around explainability, fairness, and privacy, participants consistently characterized these concerns as largely irrelevant to adoption. Explainability was not treated as a prerequisite for trust by staff at the protein processor and was sometimes viewed as potentially confusing or counterproductive. Several participants described deliberately limiting exposure to internal model representations or visualizations, noting that such information was unlikely to be meaningful to plant staff and could undermine confidence rather than build it. Instead, trust was sustained through consistent outputs, successful validation, and alignment with managerial routines.

Over time, the system receded into the background as infrastructure. The protein processor's internal project manager explained

that you'll see with some of the supervisors that joined when the system was already installed and like they just basically trusted [it] as the gospel." New supervisors and managers encountered the system as an established part of operations and accepted its outputs without revisiting earlier questions about how it worked. In this sense, trust was maintained not through ongoing scrutiny, but through the absence of disruption and the routinization of use.

## 4 Discussion

Trust literature offers multiple conceptualizations of trust in AI systems. Ribeiro et al.'s distinction between trusting individual predictions and trusting a model to behave reasonably when deployed "in the wild" provides a useful analytic lens for interpreting our findings [15]. In much machine learning practice, model trust is typically established through validation set accuracy metrics, often supplemented by inspection of individual predictions and their explanations. This framing assumes that both prediction-level scrutiny and system-level performance are central to trust formation. Our findings complicate this assumption. In the present case, organizational trust was anchored primarily at the model level rather than the prediction level. Repeatability and reproducibility were prioritized over prediction-level accuracy, and individual carcass-level outputs were neither routinely inspected nor interrogated through explanation interfaces. Instead, stakeholders accepted that individual predictions were "good enough" so long as aggregated outputs were stable and trend-sensitive over time. This suggests that organizational trust may operate at a different granularity than user-level trust, privileging system-level stability over prediction-level inspection. In this context, calibrated trust was oriented toward aggregate consistency rather than interpretability of individual decisions.

This finding also intersects with broader debates in Trustworthy AI. Many TAI frameworks position explainability and transparency as core mechanisms for enabling appropriate reliance [5], [15], [3]. However, prior research demonstrates that transparency does not uniformly increase trust; in some cases, detailed explanations can produce cognitive overload, confusion, or reduced confidence [1], [9], [20]. Explanations are typically contrastive, emerging when a particular outcome violates expectation [12]. In the present study, outputs were aggregated across ten-minute intervals rather than presented at the level of individual carcasses. By limiting granular outputs, the system reduced opportunities for contrastive questioning, thereby reducing demand for explanation.

Perceived consequence severity played a central role in shaping trust conditions. Trust in automation research has long emphasized that reliance is calibrated relative to perceived risk and potential harm [11]. Interpretability has been argued to be particularly necessary in high-stakes domains where errors carry severe consequences [16]. In this case, the system functioned in an observational role and did not directly intervene in production decisions. Its outputs were interpreted in aggregate over time rather than at the level of individual events. Errors were therefore framed as low-stakes so long as they were consistent and trend-sensitive. This low-stakes framing reduced the perceived cost of opacity and allowed trust to develop incrementally under bounded uncertainty. Future research could examine whether organizational demand for

explainability increases when AI systems become directly tied to financial incentives, compliance enforcement, or punitive evaluation mechanisms.

Moreover, while prior work suggests that users often anthropomorphize autonomous systems and attribute intentionality to their behavior [10], [19], participants in this case consistently framed the system as mechanical instrumentation rather than as an agent. Stakeholders' mental models were anchored in prior experiences with gauges, sensors, and quality control equipment rather than with autonomous decision-makers. Rather than seeking to understand why the system "decided," they sought assurance that it measured consistently within accepted tolerances. These findings do not imply that explainability is unnecessary in all contexts, but rather that its salience appears contingent on perceived agency, output granularity, and consequence severity.

Sociological studies of quantification argue that numbers derive authority not solely from technical precision, but from institutionalized validation practices and shared epistemic standards [14], [8]. In industrial settings, statistical frameworks such as gauge repeatability and reproducibility testing and ANOVA analysis function as established mechanisms for legitimizing measurement systems. In the present case, passing ANOVA testing did not resolve uncertainty about model internals; rather, it situated the system within existing regimes of industrial quality control. Statistical validation practices functioned less as technical interrogation of machine learning behavior and more as institutional rituals that conferred legitimacy.

Related research in livestock measurement technologies illustrates that ground truth itself is often socially constructed and subject to human variability [21]. Industry calibration working groups formalize acceptable error tolerances and accreditation procedures, demonstrating that measurement authority is organizationally negotiated rather than purely technical [2]. Consistent with this literature, our findings show that acceptable error thresholds were adjusted to preserve usability and alignment with existing workflows. Ground truth was not treated as a fixed empirical target but as an organizational accomplishment stabilized through practice. A value that was consistently offset yet trend-sensitive was preferable to one that was technically precise but operationally implausible.

Taken together, these findings suggest that prevailing TAI frameworks may not fully account for how trust is constructed at the organizational level in operational low-technology environments. While prior computer vision research in industrial domains frequently reports high accuracy benchmarks (e.g., sheep counting systems achieving 99.9% accuracy [21]), few studies examine how organizational actors interpret, validate, and sustain trust in these systems beyond performance metrics. Our findings indicate that accuracy metrics alone are insufficient to explain organizational trust dynamics. Instead, trust emerged through the interaction of unmet operational need, low perceived consequence severity, alignment with existing validation regimes, and routinization into infrastructure.

Rather than asking solely how to make AI systems more explainable in the abstract, designers and researchers may need to consider how AI systems can be integrated into established epistemic and organizational infrastructures. In legacy industrial contexts,

trust appears less a matter of persuasion through transparency and more a function of compatibility with existing practices of validation, oversight, and accountability. By extending trust theory from individual-level calibration to organizational-level stabilization, this study contributes an empirically grounded account of how AI becomes trusted as measurement infrastructure under conditions of bounded risk and institutional continuity.

## 5 Limitations and Future Work

This study has several limitations that shape the scope of its contributions. Although the internal project lead at the protein processor was deeply involved in the project, our data includes only a single participant from the customer organization. As a result, the findings primarily reflect trust in the system at the level of project leadership rather than across a broader range of frontline or managerial roles. In addition, this study focuses on a single customer organization and a single technology supplier. Rather than offering generalizable claims, this case is intended as a theoretically generative account of how trust in AI can emerge within a legacy industrial context.

The codebook was developed and applied primarily by one author, with iterative refinement supported through discussion with an advisor. The goal of this work is not to present a finalized coding framework but to surface analytically productive concepts that can support further collaborative refinement.

Follow-on interview studies will extend this analysis as additional systems are installed and organizational practices evolve. Future research will also incorporate system usage data, such as logs from a recently introduced verification mode, to examine how validation practices function over time as mechanisms for building, sustaining, or repairing trust in operational AI systems.

## 6 Conclusion

This paper examined how trust in an AI-based CV system developed at an organizational level in protein processing facilities through an interview study. Contrary to dominant assumptions in TAI research, trust in this specific system did not seem to emerge through explainability. Instead, trust appeared to be built through hardware reliability, familiar validation rituals, low-stakes framing of error, and the translation of AI into existing organizational categories of measurement infrastructure.

By documenting how trust was enacted through practice rather than belief in this particular system, this study contributes to trust in automation literature to contemporary AI systems deployed in legacy industrial contexts. Our findings highlight the importance of organizational need, epistemic alignment, and routinization in shaping trust, and juxtapose the explainability-centric approaches to TAI. As AI systems continue to enter operational environments far removed from their design contexts, understanding trust as an organizational and infrastructural accomplishment will be critical. This work contributes an empirically grounded account of how AI became trusted not by being understood, but by becoming ordinary.

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